

Aidan Hirsch

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EDUCATION

Purdue University
Bachelor of Science in Biomedical Engineering

3.5/4.0 GPA – May 2028
West Lafayette, IN

RESEARCH EXPERIENCE

Research Assistant

Dadarlat Lab – Purdue University

Jan 2026 – Present
West Lafayette, IN

- Modeling and labeling forelimb movements in mouse behavioral videos for DeepLabCut pose estimation model to be used in conjunction with sensorimotor cortex data for neural analysis
- Rebuilt and maintaining lab's DeepLabCut training and analysis pipeline, restoring video pose estimation for multiple active experiments
- Trained electrode-implanted mice on somatosensory cortex behavioral tasks for neural stimulation studies

Research Assistant

Neuroprostheses Research Lab – Purdue University

Jan 2025 – Dec 2025
West Lafayette, IN

- Discovered novel correlation between modeled circuit values and electrode degradation by analyzing in-vivo electrical component parameters through UMAP statistical analysis
- Analyzed ultramicroelectrode array failure modes using 16-week electrochemical impedance spectroscopy study in rat somatosensory cortex
- Engineered reusable automation pipeline for data pre-processing, measurement modeling, and file preparation, saving over 4 months of manual labor and enabling rapid statistical model iteration

Research Assistant

Friel Lab – Burke Neurological Institute

Apr 2021 – May 2024
White Plains, NY

- Designed quantitative bimanual hand assessment for children with cerebral palsy, boosting joint location accuracy by 30-50% over prior qualitative methods
- Engineered a custom drawer for rehabilitation hand assessment using Raspberry Pi and 3D cameras to capture participant joint locations in 3D space
- Developed Python program to interact with drawer assessment by determining when a participant moved or touched parts of the drawer through measurement of electrostatic discharge

LEADERSHIP AND PROJECTS

Chair

Purdue IEEE Systems Man and Cybernetics Committee

Jan 2026 – Present
West Lafayette, IN

- Leading Purdue's only neuroengineering club with 50+ active club members in development of a 128-channel EEG from scratch
- Directing Electrical, Mechanical, Software, and Research teams in their research and design goals of respective components of the final EEG system
- Conducting weekly meetings teaching various aspects of neuroscience and BCI technology developments to club members

EEG Build Team

Purdue IEEE Systems Man and Cybernetics Committee

Aug 2025 – Present
West Lafayette, IN

- Co-engineered a custom 128-electrode EEG system from component level encompassing custom ADC and microcontroller board layouts and firmware development
- Designed and fabricated multi-layer printed circuit boards to integrate 16 ADS1299 ADCs and 5 ESP32-S3 microcontrollers to achieve high fidelity and low noise neural signal acquisition
- Implemented embedded systems network using 5 ESP32s to synchronize and stream data from all 128 electrode channels at once

SKILLS

Engineering Tools: KiCad | LTSpice | PCB Design | Embedded Systems | Solidworks

Data Analysis: Python (pandas, NumPy, matplotlib, seaborn) | DeepLabCut | CEBRA | MATLAB | R

Biomedical Techniques: Impedance spectroscopy (EIS) | Rat handling (IACUC) | Cyclic Voltammetry (CV)